

Problem Sheet 8 Solutions

1.

(a) We can describe the first pulse as a superposition of two unit steps with a time-shift

$$\begin{aligned} f_1(t) &= 2u(t) - 2u(t - 1.5) \\ &= 2[u(t) - u(t - 1.5)] \\ F_1(s) &= 2[\mathcal{L}\{u(t)\} - \mathcal{L}\{u(t - 1.5)\}] \end{aligned}$$

The first term is easy, for the second we need to make use of the shift property of the LT:

$$\mathcal{L}\{f(t - \lambda)u(t - \lambda)\} = e^{-s\lambda}F(s)$$

In our case $\lambda = 1.5$ and $f(t - \lambda) = 1$.

$$\begin{aligned} F_1(s) &= 2\left(\frac{1}{s} - \frac{e^{-1.5s}}{s}\right) \\ &= \frac{2}{s}(1 - e^{-1.5s}) \end{aligned}$$

(b) The second pulse is the same as the first but delayed by 3 units of time

$$f_2(t) = f_1(t - 3)$$

Therefore,

$$F_2(s) = F_1(s)e^{-3s}$$

(c) Likewise,

$$F_3(s) = F_1(s)e^{-6s}$$

Hence the LT of the three pulses combined is

$$F(s) = F_1(s) + F_2(s) + F_3(s) = F_1(s)(1 + e^{-3s} + e^{-6s})$$

(d) If the pulses now repeat recurrently,

$$\begin{aligned} F(s) &= F_1(s)(1 + e^{-3s} + e^{-6s} + \dots) \\ &= \frac{F_1(s)}{1 - e^{-3s}} \\ &= \frac{2(1 - e^{-1.5s})}{s(1 - e^{-3s})} \\ &= \frac{2}{s} \left(\frac{1}{1 + e^{-1.5s}} \right) \end{aligned}$$

2.

(a) The steady state current at a time $t < 0$ (before the opening of the switch) is

$$i = \frac{V}{R}.$$

Since when the current is not changing, the inductor plays no part ($v = L \frac{di}{dt}$).

(b) In the limit $t \rightarrow \infty$ after the opening of the switch, likewise

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$$i = \frac{V}{R}.$$

(c) Immediately after the opening of the switch, the situation is more complex. First, we need to write down the differential equation describing the circuit. This we do by equating the voltages across the inductors and the resistor to the voltage supplied by the source V . Recall that the voltages around the loop must sum to zero (Kirchoff's voltage law). Therefore

$$(L_1 + L_2) \frac{di}{dt} + iR = V.$$

For time $t > 0$ (which is all we are interested in), we can write this as

$$(L_1 + L_2) \frac{di}{dt} + iR = Vu(t).$$

Recalling that the LT is single-sided, we now proceed to transform the differential equation into the s-domain using the linearity and first time-derivative properties:

$$I(s)s(L_1 + L_2) - i(0^+)(L_1 + L_2) + I(s)R = \frac{V}{s}$$

$$I(s)(R + s(L_1 + L_2)) = \frac{V}{s} + i(0^+)(L_1 + L_2)$$

$$I(s) = \frac{V}{s(R + s(L_1 + L_2))} + \frac{i(0^+)(L_1 + L_2)}{R + s(L_1 + L_2)}.$$

To progress further, we need an expression for $i(0^+)$, the current in the circuit just after the switch is opened. Here, $i(0^+)$ is the 'initial condition' of the circuit immediately after the discontinuity (opening the switch). Correspondingly, $i(0^-)$ is the current immediately before opening the switch, i.e. the steady-state current:

$$i(0^-) = \frac{V}{R}.$$

To find $i(0^+)$ we need to know that the flux in the inductors is conserved across the discontinuity

$$L_1 i(0^-) = (L_1 + L_2) i(0^+)$$

Hence

$$i(0^+) = \frac{VL_1}{R(L_1 + L_2)}.$$

And so

$$I(s) = \frac{V}{s(R + s(L_1 + L_2))} + \frac{\frac{VL_1}{R}}{R + s(L_1 + L_2)}.$$

Before we can inverse transform this equation, we need to simplify both terms

$$I(s) = \frac{V}{L_1 + L_2} \left[\frac{1}{s \left(s + \frac{R}{L_1 + L_2} \right)} \right] + \frac{VL_1}{R(L_1 + L_2)} \left[\frac{1}{s + \frac{R}{L_1 + L_2}} \right]$$

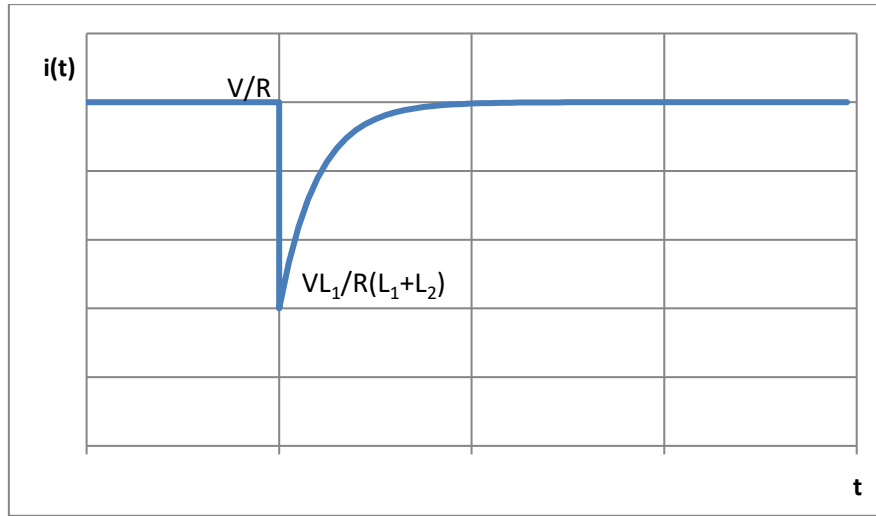
To perform the inverse transform we use entries 7 and 4 in the table of LT pairs

$$i(t) = \frac{V}{R} \left[1 - e^{-Rt/(L_1 + L_2)} \right] + \frac{VL_1}{R(L_1 + L_2)} \left[e^{-Rt/(L_1 + L_2)} \right]$$

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$$= \frac{V}{R} \left[1 - \frac{L_1}{L_1 + L_2} e^{-Rt/(L_1 + L_2)} \right]$$

The form of the current is given in the figure below



3.

(a) Marking-up the block diagram we have that the output of G_0 is $YG_0(s)$, the output of G_1 is $XG_1(s)$ and the output of G_2 is $XG_1(s)G_2(s)$. Then the output of the lower summing-point is $7X(s)G_1(s)G_2(s) - 3X(s)G_1(s)$ and we can write

$$X(s) = Y(s)G_0(s) + 7X(s)G_1(s)G_2(s) - 3X(s)G_1(s).$$

The overall transfer function is given by

$$G(s) = \frac{X(s)}{Y(s)} = \frac{G_0(s)}{1 + G_1(s)(3 - 7G_2(s))}.$$

(b) We need to find the s-domain transfer functions from the time-domain impulse responses

$$g_0(t) = 24te^{-2t} \therefore G_0(s) = \mathcal{L}\{g_0(t)\} = \frac{24}{(s+2)^2}$$

$$\mathcal{L}\left\{\frac{d}{dt}f(t)\right\} = sF(s) \therefore G_1(s) = s$$

$$\mathcal{L}\left\{\int f(t)dt\right\} = \frac{F(s)}{s} \therefore G_2(s) = \frac{1}{s}$$

$$G(s) = \frac{\frac{24}{(s+2)^2}}{1 + s\left(3 - \frac{7}{s}\right)} = \frac{24}{(s+2)^2(3s-6)}.$$

(c) $G(s) \rightarrow \infty$ for $s = \pm 2$ where $s = \sigma + j\omega$. The roots are therefore real, so the system will not oscillate (i.e. no sinusoids). Roots on the LHS of the imaginary axis ($\sigma < 0$) are stable. However, any single root on the RHS results in instability, i.e. a small signal at the input will result in continuous growth at the output. Therefore this system is **not** stable.

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(d) For $y(t) = u(t)$ we have $Y(s) = \frac{1}{s}$ so we can write

$$X(s) = G(s)Y(s) = \frac{24}{s(s+2)^2(3s-6)} = \frac{8}{s(s+2)^2(s-2)}$$

Proceed as usual using partial fractions

$$\frac{8}{s(s+2)^2(s-2)} = \frac{A}{s} + \frac{B}{s+2} + \frac{C}{(s+2)^2} + \frac{D}{s-2}$$

$$\left| \frac{8}{(s+2)^2(s-2)} \right|_{s=0} \xrightarrow{\text{yields}} A = -1$$

$$\left| \frac{8}{s(s+2)^2} \right|_{s=2} \xrightarrow{\text{yields}} D = \frac{1}{4}$$

$$\left| \frac{8}{s(s-2)} \right|_{s=-2} \xrightarrow{\text{yields}} C = 1$$

We cannot use $s = -2$ again directly to get B . We could differentiate the above expression with respect to s and then use $s = -2$. However, as we now have only one unknown left in the partial fraction we can choose another value say $s = 3$ and evaluate

$$\left| \frac{8}{s(s+2)^2(s-2)} \right|_{s=3} \xrightarrow{\text{yields}} B = \frac{3}{4}$$

Therefore

$$X(s) = -\frac{1}{s} + \frac{3}{4(s+2)} + \frac{1}{(s+2)^2} + \frac{1}{4(s-2)}$$

$$\mathcal{L}^{-1}\left\{-\frac{1}{s}\right\} = -u(t) = -1 \text{ for } t > 0$$

$$\mathcal{L}\left\{\frac{1}{4(s-2)}\right\} = \frac{e^{2t}}{4}$$

$$\mathcal{L}\left\{\frac{3}{4(s+2)}\right\} = \frac{3e^{-2t}}{4}$$

$$\mathcal{L}\left\{\frac{1}{(s+2)^2}\right\} = te^{-2t}$$

Hence

$$x(t) = -1 + \frac{e^{2t}}{4} + \frac{3e^{-2t}}{4} + te^{-2t}$$

For all $t > 0$

Note:

We see the instability due to the e^{2t} term.